

PROVISIONAL PATENT APPLICATION

Multi-Stage Counter-Flow Fluidized Bed Heat Exchange System for Extraterrestrial Regolith Processing

Project Designation: RCFX (Regolith Counter-Flow Exchange)

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1. Background of the Invention

This invention relates generally to heat recovery systems for in-situ resource utilization (ISRU) processing of extraterrestrial regolith. More specifically, it relates to a multi-stage counter-current fluidized bed heat exchanger operating within a pressurized envelope maintained by process-derived volatile gases, designed for use in reduced-gravity vacuum environments such as the lunar surface, Mars, or asteroidal bodies.

ISRU processes including Molten Regolith Electrolysis (MRE) and hydrogen reduction require heating regolith to 900 to 1900 K. Without heat recovery, the spent regolith is discharged at extraction temperature and its thermal energy is lost. For a 100 kg/hr MRE facility processing regolith from 200 K ambient to 1600 K extraction temperature, the total thermal budget including sensible heating (31 kW), latent heat of fusion (12.5 kW), and reactor thermal losses is approximately 70 kW. For hydrogen reduction at 900 K, the sensible heating requirement alone is approximately 16 kW. The disclosed heat recovery system operates over the 200 to 900 K range, recovering sensible thermal energy that would otherwise be wasted when spent regolith is discharged. At 80 percent recovery effectiveness, this saves approximately 12 to 16 kW of sensible heating power at pilot scale. For MRE operations, the total system savings including reduced reactor thermal losses from receiving preheated feed are approximately 25 kW. These savings scale linearly with throughput.

Extraterrestrial environments defeat standard heat exchanger designs. The lunar surface is hard vacuum (below $1\text{E-}12$ torr), which eliminates convective heat transfer without an engineered atmosphere. Mars offers a thin CO₂ atmosphere at approximately 6 mbar, which is insufficient for effective convective exchange at ISRU-relevant temperatures and flow rates. Asteroidal surfaces present microgravity conditions that further complicate granular material handling. In all cases, regolith is extremely abrasive, composed of angular silicate particles with Mohs hardness of 5 to 7.

Lunar regolith is electrostatically charged through triboelectric and UV-photoemission processes to kilovolt potentials. Apollo samples show a median grain size of 60 to 80 micrometers with 10 to 20 percent below 20 micrometers. This bimodal distribution spans the Geldart A/C boundary, meaning the fine fraction is cohesive and dominated by electrostatic and van der Waals forces rather than gravitational settling. Mars regolith exhibits similar characteristics with additional perchlorate chemistry concerns. No existing ISRU heat recovery system has been demonstrated that addresses the vacuum, abrasion, electrostatic, clogging, cohesion, and energy-waste problems simultaneously.

The disclosed system addresses six specific challenges that collectively define the extraterrestrial regolith heat recovery problem. First, vacuum eliminates convective heat transfer. Second, regolith is extremely abrasive. Third, regolith is electrostatically charged. Fourth, fine particles cause clogging in fixed-geometry exchangers. Fifth, the bimodal particle size distribution includes a cohesive fraction that resists fluidization. Sixth, the thermal energy at stake (16 to 70 kW continuous for a pilot-scale facility depending on extraction process) represents a substantial fraction of total mission power.

2. Summary of the Invention

The disclosed system is a multi-stage counter-current fluidized bed heat recovery system operating within a pressurized envelope maintained by process-derived volatile gases. It is analytically estimated to achieve greater than 70 percent thermal energy recovery (design target 80 to 90 percent) with total parasitic power of 80 to 250 watts, which is less than 2 percent of the 12 to 16 kW of sensible heat being recovered at pilot scale. Pilot-scale mass is approximately 470 kg at 50 to 100 kg/hr throughput.

The system uses counter-current staged fluidized beds as the heat exchange mechanism. Fluidized beds provide vigorous mixing that prevents clogging and ensures uniform heat transfer. The beds operate within a pressurized processing envelope at 0.2 to 0.3 bar (design range 0.1 to 0.5 bar) using volatile gases (H₂, He, CO, CO₂, N₂) extracted from the regolith itself, enabling convective heat transfer in vacuum. The gas inventory is self-sustaining with a generation-to-loss ratio exceeding 50:1. Each stage contains iron-based thermal mass (1 to 5 mm spheres produced from MRE byproduct) to buffer thermal energy, prevent sintering, and disrupt cohesive agglomerates.

Nine key innovations define the system. First, counter-flow staged fluidized beds for regolith heat recovery. Second, a pressurized volatile envelope enabling convection in vacuum. Third, iron-based thermal mass from extraction byproduct with in-situ surface hardening. Fourth, Electrodynamic Dust Shield (EDS) integration within high-temperature fluidized bed stages. Fifth, rotary airlock material transfer with gas purge isolation. Sixth, sintering prevention through iron shot tumbling action. Seventh, forced gas circulation using a compact low-power blower with N+1 redundancy and passive thermosiphon limp-home mixing. Eighth, active particle size management for the bimodal Geldart A/C transition characteristic of extraterrestrial regolith. Ninth, staged bootstrap deployment enabling progressive transition from Earth-supplied to locally-produced and metallurgically optimized iron thermal mass.

The thermal recovery effectiveness is derived from the counter-flow energy balance. Effectiveness is defined as the ratio of actual cold-stream temperature gain to the maximum possible temperature gain: $\epsilon = (T_{\text{cold_out}} - T_{\text{cold_in}}) / (T_{\text{hot_in}} - T_{\text{cold_in}})$. With $T_{\text{cold_in}}$ equal to 200 K, $T_{\text{cold_out}}$ equal to 760 K, and $T_{\text{hot_in}}$ equal to 900 K, the effectiveness is $(760 - 200) / (900 - 200)$, which equals 560 divided by 700, yielding ϵ equal to 0.80. The corresponding energy recovered is calculated as $Q_{\text{recovered}} = \text{mass flow rate} \times \text{specific heat capacity} \times (T_{\text{hot_in}} - T_{\text{hot_out}})$, where $T_{\text{hot_out}}$ equals 340 K. Substituting: $Q_{\text{recovered}}$ equals 0.0278 kg/s times 800 J/kg/K times $(900 - 340)$ K, yielding $Q_{\text{recovered}}$ approximately equal to 12,400

W or 12.4 kW. This derivation anchors the 80 percent recovery claim at pilot-scale throughput of 100 kg/hr. The design target of 80 to 90 percent effectiveness is pending Phase 1 modeling validation of heat transfer coefficients, gas-solid coupling, and residence time distribution across the five stages.

3. Brief Description of the Drawings

FIG. 1 is a schematic overview of the RCFX system showing the five-stage counter-flow arrangement within the pressurized envelope, with cold regolith entering at Stage 1 and hot spent regolith returning from the extraction reactor through Stage 5. Reference numerals: 100 (complete RCFX system), 110 (pressurized envelope), 120a through 120e (Stages 1 through 5), 130 (gas circulation manifold), 140 and 142 (inlet and outlet rotary airlocks), 150 (EDS electrode arrays).

FIG. 2 is a cross-sectional view of a single fluidized bed stage showing the internal arrangement of the sintered metal distributor plate, iron shot thermal mass bed, EDS electrode arrays on vessel walls, alumina ceramic liners, and freeboard region. Reference numerals: 122 (stage vessel), 124 (alumina liner), 126 (distributor plate), 128 (iron shot thermal mass), 129 (freeboard region), 150 (EDS electrodes), 160 (regolith flow path).

FIG. 3 is a gas circulation schematic showing the forced circulation loop driven by blower 132a (primary) and 132b (redundant standby), with the pressurized envelope 112, manifold connections to each stage, cyclone separator 170, and the quantitative loop resistance breakdown. The stages are connected in parallel to the supply and return manifolds, so the blower overcomes the pressure drop across a single stage (approximately 7,000 Pa) rather than all five in series. The thermosiphon buoyancy contribution is indicated as supplementary passive mixing only, with the deficit between buoyancy driving pressure (0.06 Pa) and per-stage loop resistance annotated.

FIG. 4 is a temperature profile diagram showing the counter-flow heat exchange across all five stages, with cold regolith temperatures on the ascending path and hot spent regolith temperatures on the descending path, demonstrating the approximately 140 K temperature differential per stage.

FIG. 5 is a detailed view of the rotary airlock assembly showing the four-phase transfer cycle (A: fill, B: pressurize/purge, C: discharge, D: vent/reset), ceramic wear liners, and gas purge ports.

FIG. 6 is an integration schematic showing (a) RCFX coupled with a Molten Regolith Electrolysis reactor including the iron shot thermal mass feedback loop and staged iron replacement pathway, and (b) RCFX coupled with a hydrogen reduction reactor including the volatile gas feedback loop.

4. Detailed Description of the Preferred Embodiment

4.1 System Overview

Referring to FIG. 1, the RCFX system 100 comprises a pressurized processing envelope 110 containing five fluidized bed heat exchange stages 120a through 120e arranged in counter-current configuration. Cold incoming regolith enters at Stage 1 (120a) at approximately 200 K through inlet airlock 140 and progresses through stages of increasing temperature.

Simultaneously, hot spent regolith from the extraction reactor enters at Stage 5 (120e) at approximately 900 K through outlet airlock 142 and progresses through stages of decreasing temperature. For extraction processes operating above 900 K (such as MRE at 1600 K), a pre-cooling stage or radiative dump reduces the spent regolith temperature to approximately 900 K before entry into the RCFX, or the system may be designed with additional high-temperature stages to accommodate the higher entry temperature. The two regolith streams exchange thermal energy through the intermediate iron shot thermal mass and gas-phase convection within each stage.

The five-stage arrangement provides temperature steps of approximately 100 to 140 K per stage, which limits thermal shock to regolith particles and structural components while maintaining high overall recovery efficiency. The stage count represents an engineering optimum: fewer stages reduce recovery efficiency below the 70 percent minimum, while additional stages add mass and complexity with diminishing thermal returns.

4.2 Pressurized Processing Envelope

The pressurized processing envelope 110 is maintained at 0.2 to 0.3 bar nominal operating pressure (design range 0.1 to 0.5 bar) using volatile gases extracted from the regolith during processing. The envelope serves four critical functions. First, it enables convective heat transfer that is physically impossible in vacuum. Second, it suppresses electrostatic charging of regolith particles by providing gas-phase ion discharge paths. Third, it contains fine particles within the processing volume. Fourth, it provides the working fluid for fluidization of the bed stages.

The envelope is constructed of Ti-6Al-4V pressure vessel rated for 1.0 bar maximum allowable working pressure with a 2:1 safety factor over nominal operating pressure. Internal surfaces exposed to regolith contact are lined with sacrificial alumina ceramic liners 124 (greater than 99.9 percent purity Al₂O₃, specifically excluding transition metal dopants such as chromium or manganese that increase high-temperature leakage currents via altered trap densities, 5 mm nominal thickness) to protect the structural pressure boundary from abrasion and provide dielectric insulation for the EDS electrode arrays.

The exterior of the pressure vessel is wrapped in multilayer insulation (MLI) to minimize radiative heat loss to the vacuum environment. At the operating temperature range of 200 to 900 K, an uninsulated pressure vessel with approximately 4 m² of external surface area and average shell temperature of 600 K would radiate approximately 24 kW to space (calculated as emissivity 0.8 times Stefan-Boltzmann constant times 600 K to the fourth power times 4 m²). This exceeds the sensible heat being recovered and would render the system thermally non-viable. MLI reduces the effective emissivity to 0.02 to 0.05, bringing external radiative losses to approximately 0.6 to 1.5 kW, which is negligible relative to system thermal flows. Internal radiative exchange between stages within the pressurized envelope is beneficial rather than parasitic: radiation from hotter stages to cooler stages supplements the convective counter-flow gradient and is captured within the thermal circuit. Internal radiation shields or staggered stage geometry may be employed to prevent direct radiative bypass from Stage 5 to Stage 1 if view factor analysis during Phase 1 modeling indicates significant thermal short-circuiting.

4.3 Gas Inventory, Volatile Liberation, and Self-Sustaining Operation

The gas inventory within the pressurized envelope is self-sustaining under steady-state operation. Lunar regolith contains approximately 50 to 100 ppm of solar-wind-implanted volatile species (primarily H₂, He, and N₂). However, when heated to extraction temperatures (700 to 1600 K), additional volatiles are released through thermal decomposition of carbonaceous compounds and incipient reduction of mineral oxides, yielding a total recoverable volatile fraction of approximately 200 to 300 ppm by mass. At 100 kg/hr regolith throughput, this yields approximately 20 to 30 grams per hour of volatile gas liberation. The total gas inventory required to maintain the envelope at 0.2 bar is approximately 15 grams (depending on envelope volume and average gas temperature).

The volatile gases are liberated in a characteristic three-tier thermal desorption sequence as regolith progresses through the counter-flow stages. In the low-temperature regime (200 to 600 K, corresponding to Stages 1 and 2), adsorbed water vapor, molecular hydrogen, and trace nitrogen are the primary species released. In the mid-temperature regime (600 to 900 K, corresponding to Stages 3 and 4), a near-quantitative release of solar-wind-implanted hydrogen and helium isotopes occurs. In the high-temperature regime (900 to 1400 K, corresponding to Stage 5 and the extraction reactor), thermal decomposition of carbonaceous compounds and incipient reduction of iron oxides by native hydrogen yields significant quantities of carbon monoxide, carbon dioxide, and additional water vapor. This desorption profile is significant because the heavier molecular species (CO, CO₂) that most benefit fluidization performance through increased gas density and aerodynamic drag are preferentially generated at the hot end of

the system, precisely where they are most needed to maintain stable fluidization of the coarsest and hottest regolith fraction.

Water vapor requires active management within the gas circulation loop. At the operating pressure of 0.2 bar, the saturation temperature of water is approximately 333 K (60 degrees Celsius). In Stages 1 and 2, where gas temperatures may drop below this threshold, water vapor would condense on distributor plates and vessel surfaces, potentially causing ice formation at the coldest points and hydrolytic degradation of the alumina ceramic liners over time. To prevent this, a cold trap or condenser is incorporated in the gas return manifold between the cold stages and the blower inlet. The cold trap removes water vapor from the circulating gas stream before it can reach condensation temperatures. The collected water is a useful ISRU byproduct that may be electrolyzed for oxygen production or stored for life support. For lunar regolith with very low total water content (typically below 50 ppm), the water vapor partial pressure may remain below the condensation threshold at all stage temperatures, in which case the cold trap operates passively as a precautionary measure rather than an active necessity. Mars regolith with higher bound water content would require more aggressive water removal.

Gas leakage through airlock seals and vessel penetrations is estimated at 0.4 grams per hour. This estimate is physically grounded in Knudsen flow regime analysis: because the external environment is hard vacuum, gas transport through micro-leaks in seals and penetrations is governed by molecular flow rather than viscous flow, strictly limited by the mean free path of the gas molecules. The generation-to-loss ratio is therefore 20 g/hr generation divided by 0.4 g/hr leakage, yielding 50:1. Even assuming seal degradation doubles the leakage rate to 0.8 g/hr over the system lifetime, the ratio remains 25:1, still providing a large margin. The steady-state gas inventory of 15 grams represents less than one hour of volatile generation, meaning the envelope reaches operating pressure within the first hour of regolith processing. A pre-charge gas bottle (0.1 kg compressed He or H₂) is included in the mass budget to provide initial pressurization during system startup before regolith processing generates sufficient volatiles.

Mars regolith contains substantially higher volatile content (including adsorbed CO₂ and bound water in hydrated minerals), which further improves the generation-to-loss margin. Asteroidal regolith, particularly carbonaceous chondrite material, may contain several weight percent volatiles, making the self-sustaining gas inventory even more robust in those environments.

4.4 Regolith Particle Size Distribution and Dust Physics

The fluidization design is grounded in the well-characterized particle size distribution of Apollo lunar regolith samples (Carrier, 2003). The representative mass fractions are: less than 20 micrometers, 10 to 20 percent; 20 to 50 micrometers, 15 to 25 percent; 50 to 100 micrometers,

15 to 25 percent; 100 to 250 micrometers, 20 to 30 percent; 250 to 500 micrometers, 10 to 15 percent; greater than 500 micrometers, small fraction. The median particle diameter is approximately 60 to 80 micrometers with a bulk grain density of approximately 3100 kg/m³.

This distribution is critical because fluidization behavior is dominated by the fine fraction, not the median. Using the Wen-Yu minimum fluidization correlation at 0.2 bar pressure with He-dominant gas and lunar gravity (1.625 m/s²), the minimum fluidization velocity (U_{mf}) for representative particle sizes is: 30 micrometers, approximately 0.2 mm/s; 70 micrometers, approximately 0.9 mm/s; 200 micrometers, approximately 6 mm/s. These are comfortably achievable velocities, confirming that fluidization is physically feasible at low pressure in reduced gravity.

The actual engineering challenge is not achieving U_{mf} but maintaining stable fluidization with a bimodal particle size distribution that spans the Geldart A/C transition boundary. The 10 to 25 percent mass fraction below 40 to 50 micrometers is in the Geldart C (cohesive) regime where electrostatic and van der Waals interparticle forces dominate gravitational settling. In vacuum, these forces produce electrostatically cemented agglomerates ('fairy castle structures') that resist fluidization entirely. The pressurized gas envelope directly mitigates this by providing ion discharge paths that reduce surface charge accumulation by orders of magnitude compared to vacuum conditions.

Mars regolith, as characterized by rover and lander missions, exhibits a finer median grain size (approximately 40 to 60 micrometers) with a higher proportion of sub-20 micrometer particles. The system design envelope of 0.1 to 0.5 bar operating pressure and the fines management architecture described in Section 5.8 accommodate this variation.

4.5 Fluidized Bed Stage Architecture

Referring to FIG. 2, each fluidized bed stage 120 comprises a cylindrical vessel 122 with internal alumina ceramic liner 124, a sintered metal distributor plate 126 at the base, iron-based thermal mass 128 distributed throughout the bed, EDS electrode arrays 150 on all interior wall surfaces, and a freeboard region 129 above the bed surface. For the pilot-scale system at 50 to 100 kg/hr throughput, each stage vessel has a nominal internal diameter of approximately 0.36 m (cross-sectional area approximately 0.1 m²) and a bed height of approximately 0.3 m, with an additional 0.3 to 0.5 m of freeboard above the bed surface for disengagement of elutriated particles. These dimensions are derived from the fluidization velocity requirements and regolith residence time needed for effective heat transfer.

The gas properties within the pressurized envelope vary with temperature and composition. For the He-dominant gas mixture at representative conditions (0.2 bar, 600 K average), the dynamic viscosity is approximately $2.5\text{E-}5$ Pa-s and the gas density is approximately 0.02 kg/m³, calculated from the ideal gas law. The specific heat capacity of lunar regolith is taken as 800 J/kg/K, a representative value for silicate minerals at the operating temperature range (published Apollo sample measurements range from 700 to 900 J/kg/K depending on composition and temperature).

The distributor plate 126 is fabricated from sintered Inconel 625 with a pore size of 20 to 30 micrometers and a thickness of 3 to 5 mm. The pore size is selected to prevent iron shot passage while allowing gas flow with acceptable pressure drop. Standard fluidized bed design practice requires the distributor pressure drop to be at least 20 to 30 percent of the bed pressure drop to ensure uniform gas distribution. In the RCFX system, quantitative analysis using the Ergun equation confirms that at 3 times U_{mf} operating velocity, the distributor accounts for approximately 94 percent of per-stage pressure drop (approximately $6,300$ Pa versus approximately 400 Pa for the bed), far exceeding the minimum uniformity requirement. This dominance is not a design deficiency. It is a consequence of the very low gas density at 0.2 bar passing through fine sintered pores, and it ensures excellent distribution uniformity. It is also the primary reason that buoyancy-driven circulation is fundamentally incompatible with properly designed fluidized bed systems at low gas density.

Regolith transfers between stages via gravity-assisted overflow weirs or controlled pneumatic transfer, with no mechanical contact surfaces in the regolith flow path. This eliminates the primary abrasion failure mode that has defeated prior regolith handling concepts.

4.6 Iron-Based Thermal Mass: Function, Degradation, and In-Situ Hardening

Each stage contains iron-based thermal mass 128 comprising spherical iron or iron alloy particles of 1 to 5 mm diameter. In the preferred embodiment, the iron shot is produced as a byproduct of Molten Regolith Electrolysis, creating a closed material loop where the extraction process generates its own thermal buffer material. For installations using hydrogen reduction or other extraction processes that do not produce metallic iron, iron shot may be supplied from Earth (estimated at 50 kg total for a pilot-scale system) or produced from locally refined ilmenite.

Iron shot provides high volumetric heat capacity (3.5 MJ/m³/K) and mechanical durability. The tumbling action of iron shot within the fluidized bed performs two additional functions: it breaks up incipient agglomerates of cohesive fine particles, and it distributes abrasion wear uniformly rather than concentrating it on fixed surfaces. The iron shot also acts as a thermal flywheel, buffering transient variations in regolith feed rate or temperature.

The iron thermal mass is a consumable component subject to continuous abrasive wear. The hardness mismatch between iron (approximately 200 HV for pure iron) and lunar silicate minerals (500 to 1100 HV for pyroxene, olivine, and quartz respectively) places the abrasive-to-metal hardness ratio (H_a/H_m) between 2.5 and 5.5, well above the critical 1.2 threshold at which three-body abrasion transitions from mild to severe. Based on industrial analogs from mineral processing ball mills and fluidized bed combustors, the estimated iron shot consumption rate is 30 to 60 grams per ton of regolith processed, or approximately 3 to 6 grams per hour at 100 kg/hr throughput. At 8,760 operating hours per year, annual replacement demand is approximately 26 to 53 kg. The MRE byproduct stream produces approximately 10 kg/hr of metallic iron (from the approximately 13 weight percent FeO content of typical highland regolith), exceeding the replacement demand by a factor of approximately 1,000 to 1.

The primary degradation pathway is progressive surface erosion and gradual diameter reduction, not catastrophic fragmentation. Individual spheres gradually round and shrink over hundreds to thousands of operating hours. Worn iron shot that shrinks below a minimum functional diameter (approximately 0.5 to 1 mm) is removed from the beds by elutriation or screening at stage boundaries, and fresh shot is added from the replacement inventory via a gravity-feed hopper at the highest-temperature stage where wear is most aggressive.

The iron wear products (iron fines in the 1 to 50 micrometer range) are generated continuously during operation. Because iron fines are substantially denser than regolith fines (7,870 versus 3,100 kg/m³), they have higher terminal velocities at equivalent particle diameter and are therefore easier to capture cyclonically than equivalent-size regolith particles. Magnetic separation may be employed as a secondary fines capture mechanism, leveraging the ferromagnetic properties of the iron fines to separate them from the non-magnetic regolith discharge stream. Recovered iron fines may be collected for re-melting and re-granulation into replacement shot.

A significant and novel feature of the RCFX architecture is that the operating environment provides in-situ surface hardening of the iron thermal mass at no additional cost. The pressurized gas envelope contains carbon monoxide as a major component (Section 4.3), and the operating temperature of Stages 3 through 5 (600 to 900+ K) places the iron shot within the thermodynamic window for gas carburization. Carbon diffuses from the CO atmosphere into the iron surface, forming a case of iron carbide (Fe_3C , cementite) with hardness of 800 to 1000 HV. This reduces the H_a/H_m ratio against quartz from 5.5 down to approximately 1.1 to 1.4, at or near the critical transition threshold between severe microcutting and moderate wear. Against olivine and pyroxene, the ratio drops below 1.2 into the mild wear regime entirely. Case depth at 900 K in CO atmosphere is estimated at 0.1 to 0.3 mm over hundreds of hours of residence time,

providing meaningful protection for 1 to 5 mm spheres. This passive carburization requires no additional processing, materials, energy, or hardware. It occurs as a natural thermochemical consequence of the CO-enriched gas environment specified for fluidization performance in Section 4.3 and Claim 27.

4.7 Staged Iron Shot Deployment and Bootstrap Sequence

The iron thermal mass deployment follows a staged bootstrap sequence consistent with the incremental nature of ISRU facility commissioning. In the initial deployment (Stage 0), the system lands with approximately 50 kg of Earth-manufactured iron shot optimized for the application: precisely sized, heat-treated to 40 to 50 HRC, with known metallurgy. This seed stock provides reliable performance during system commissioning when all other subsystems are also being validated.

Once the MRE extraction reactor achieves steady-state operation and begins producing metallic iron as a byproduct, the first locally-produced replacement shot (Stage 1) is introduced. This first-generation on-site shot is pure iron at approximately 200 HV, granulated by controlled quenching of tapped molten iron. It wears at the maximum rate (26 to 53 kg/year) but is produced in vast excess of demand.

As operational experience with the MRE accumulates, the iron melt composition can be deliberately managed (Stage 2). By controlling melt residence time and electrode chemistry, 0.3 to 0.8 percent carbon can be dissolved into the molten iron before tapping. Moderate quenching produces pearlitic or bainitic iron shot at 350 to 500 HV, approximately doubling the service life relative to pure iron with no additional hardware or consumables.

The final optimization (Stage 3) occurs passively as the CO-enriched gas envelope carburizes the surface of all iron shot in the beds over hundreds of hours of operation. The result is a case-hardened pearlitic core with a cementite surface layer at 800 to 1000 HV. At this stage, annual iron shot replacement is estimated at 5 to 15 kg/year, and the MRE iron output exceeds consumption by approximately 1,000 to 1. The system has transitioned from imported consumable to self-sustaining closed-loop material cycle.

5. Key Innovations

5.1 Counter-Flow Staged Fluidized Beds

The first innovation is the use of counter-flow staged fluidized beds as the primary heat recovery mechanism for ISRU regolith processing. Fluidized beds provide vigorous gas-solid mixing that eliminates the channeling and clogging failure modes that have defeated fixed-geometry heat exchangers (including packed beds, honeycomb channels, and concentric tube designs) when handling angular, cohesive, electrostatically charged regolith. The counter-flow arrangement maximizes the temperature driving force across each stage while limiting per-stage thermal shock.

5.2 Pressurized Volatile Envelope

The second innovation is the pressurized processing envelope maintained by volatile gases extracted from the regolith itself. This converts the vacuum environment from a liability (no convection possible) into a design parameter (controlled gas composition and pressure optimized for fluidization). The gas medium provides three simultaneous functions: convective heat transfer, electrostatic charge dissipation, and fluidization of the bed stages. The self-sustaining gas inventory (generation-to-loss ratio exceeding 50:1) eliminates consumable requirements. The temperature-dependent volatile liberation profile further benefits the system by delivering heavier CO and CO₂ species preferentially at the hot end where gas density matters most for stable fluidization.

5.3 Iron-Based Thermal Mass with In-Situ Surface Hardening

The third innovation is the use of MRE-derived iron shot as thermal mass within each fluidized bed stage, combined with passive in-situ surface carburization by the carbon monoxide component of the pressurized gas envelope. This creates a closed-loop material synergy between the extraction reactor and the heat recovery system, where a waste product of the extraction process becomes a critical functional component of the heat exchanger that progressively improves in service through thermochemical hardening. The iron shot simultaneously serves as thermal buffer, agglomerate breaker, sintering preventer, and abrasion distributor, with surface hardness increasing from approximately 200 HV (as-produced pure iron) to 800 to 1000 HV (carburized surface) during normal operation.

5.4 EDS Integration in High-Temperature Fluidized Beds

The fourth innovation is the integration of Electrodynamic Dust Shield electrode arrays within high-temperature fluidized bed stages for active dust mitigation. EDS technology has been

demonstrated for surface dust removal in ambient conditions but has not been previously applied inside operating fluidized beds at temperatures up to 1000 K. The EDS electrodes prevent charged fine particles from accumulating on vessel walls where they would form cohesive layers that grow into flow obstructions. In a no-maintenance extraterrestrial environment, wall fouling is a terminal failure mode.

Together, the EDS electrodes, the pressurized envelope, and the fluidized bed mixing action form an interlocking defense against the specific dust physics failure mode that would defeat any single mitigation approach used in isolation.

The EDS electrodes use high-temperature ceramic-insulated wiring embedded within or beneath the sacrificial alumina liners. The alumina liners are specified at greater than 99.9 percent purity, specifically excluding transition metal dopants (chromium, manganese) that increase high-temperature leakage currents via Frenkel-Poole emission through altered trap densities in the crystal lattice. At operating temperatures approaching 1000 K, the volume resistivity of alumina decreases by several orders of magnitude from room-temperature values; the high-purity specification ensures that the reduced resistivity remains adequate for dielectric isolation of the EDS operating voltages (1 to 4 kV). The EDS operating voltage must be characterized during Phase 2 testing against the Paschen breakdown curve for the specific gas composition at 0.2 bar operating pressure to ensure adequate margin against micro-arcing through liner defects. Ground fault detection on each EDS phase provides continuous liner integrity monitoring. Power consumption is approximately 1 to 5 watts per square meter of protected surface area, with a total system budget of 5 to 15 watts.

5.5 Rotary Airlock Material Transfer

The fifth innovation is the rotary airlock assembly with gas purge isolation for material transfer between the vacuum exterior and the pressurized interior. Each airlock operates a four-phase cycle: fill (receive regolith from hopper or previous stage), pressurize/purge (inject gas to match envelope pressure and flush entrained vacuum), discharge (release regolith into pressurized envelope), and vent/reset (controlled depressurization and return to fill position). Ceramic wear liners protect all regolith-contact surfaces within the airlock.

NASA empirical testing of rotary seals designed for lunar environments, using JSC-1A and LHT-2M simulants in vacuum chambers through 10,000-cycle endurance tests, has demonstrated that PTFE-based seals can successfully exclude simulant from the seal-shaft interface. The abrasive action causes the PTFE to flake and transfer onto the counter-face, creating a self-lubricating tribological film that stabilizes wear rates over time rather than allowing monotonically increasing degradation. This data supports the feasibility of the rotary airlock seal

concept, though the RCFX requires continuous operation over months to years at throughputs of 876 metric tonnes per year, substantially exceeding the validated cycle count. The architecture may evolve to screw conveyors or diaphragm valves during Phase 2 bench testing. The 50:1 gas generation-to-loss margin provides the necessary headroom to accommodate progressive seal degradation.

5.6 Sintering Prevention

The sixth innovation is sintering prevention through the tumbling action of iron shot within the fluidized bed. At temperatures approaching 900 to 1000 K, regolith particles become susceptible to solid-state sintering at contact points, which would fuse particles into immobile masses and kill the fluidized bed. The continuous tumbling and impact of iron shot spheres against regolith particles disrupts incipient sinter bonds before they can consolidate, maintaining bed fluidity without mechanical stirring mechanisms that would be subject to rapid abrasive failure.

5.7 Forced Gas Circulation with Passive Backup

The seventh innovation is forced gas circulation using a compact low-power blower sized to maintain superficial gas velocity at 3 to 5 times the minimum fluidization velocity for the target particle size fraction. The five fluidized bed stages are connected in parallel to supply and return gas manifolds, so the blower overcomes the pressure drop across a single stage (approximately 7,000 Pa, dominated by the sintered metal distributor plate) rather than all five in series. The blower is rated at 70 to 150 watts depending on operating velocity, gas composition, and blower efficiency, representing less than 2 percent of the sensible heat being recovered. Quantitative loop analysis establishes that thermosiphon buoyancy from the 340 to 900 K temperature gradient is approximately five orders of magnitude below the per-stage loop resistance at operating velocities, a direct consequence of reduced gravity and very low gas density at 0.2 bar. The thermosiphon effect therefore serves only as a passive limp-home mixing mode that slows thermal gradient collapse during blower off-states, providing additional fault recovery time without achieving fluidization.

The system employs N+1 blower redundancy with a primary blower 132a and standby blower 132b. Automatic crossover on RPM or differential pressure fault occurs within 2 seconds. Blowers use ceramic bearings and brushless motors for longevity in the dusty, high-temperature environment.

5.8 Particle Size Management and Fines Control

The eighth innovation addresses the bimodal Geldart A/C transition character of extraterrestrial regolith through active fines management. At the superficial gas velocities required for effective

fluidization of the target size fraction (approximately 200 micrometers), which are typically 3 to 5 times the minimum fluidization velocity, particles below approximately 30 to 40 micrometers have terminal velocities below the operating gas velocity and are therefore elutriated in the gas stream. Quantitative analysis using the Haider-Levenspiel terminal velocity correlation for non-spherical particles (sphericity 0.7 for angular regolith) at 0.1 bar with helium-dominant gas confirms that the entrainment cutoff lies at approximately 37 micrometers across all five stages at 3 times the minimum fluidization velocity of the 200-micrometer fraction.

The system addresses fines entrainment through three complementary approaches, which may be employed individually or in combination depending on specific mission requirements and feedstock characteristics.

The first approach is cyclone separation. One or more cyclone separators are installed in the gas circulation loop downstream of each stage or at consolidated points in the gas manifold. Elutriated fine particles are captured by centrifugal separation and either returned to the fluidized bed for heat recovery or diverted to a separate fines collection reservoir for alternative processing. Iron fines from thermal mass degradation are denser than regolith fines and are therefore captured with higher efficiency at equivalent particle diameter. Magnetic separation may be integrated as a secondary capture mechanism for ferromagnetic iron fines.

The second approach is pre-classification. A sieve or screening stage upstream of the RCFX system removes the Geldart C fraction (below 40 to 50 micrometers) before it enters the heat exchanger. The separated fines bypass the fluidized beds and are either processed separately or blended back into the output stream after heat exchange. This approach reduces the cohesive particle loading in the beds at the cost of reduced heat recovery from the fines fraction.

The third approach is acceptance of fines loss with the understanding that elutriated material is captured in the gas loop and periodically purged. This approach is appropriate when the fines fraction represents a small percentage of total feed mass and the economic impact of bypassing their heat recovery is acceptable.

5.9 Staged Bootstrap Deployment of Iron Thermal Mass

The ninth innovation is the staged deployment strategy for iron thermal mass, wherein Earth-manufactured seed stock is progressively replaced by locally-produced iron with improving metallurgical properties. The initial Earth-supplied seed stock (approximately 50 kg) provides optimal performance during system commissioning. As MRE operations produce iron byproduct, locally-granulated pure iron replaces the consumed seed stock. Progressive improvements to the locally-produced iron are achieved through controlled carbon dissolution during electrolysis

(yielding pearlitic iron at 350 to 500 HV) and passive surface carburization by the CO-enriched gas envelope (yielding cementite surface layers at 800 to 1000 HV). This staged approach reduces the annual iron replacement requirement from an initial 26 to 53 kg/year (pure iron) to an estimated 5 to 15 kg/year (carburized pearlitic iron) while transitioning from an Earth-dependent consumable to a fully self-sustaining closed-loop material cycle. The deployment sequence is inherently fault-tolerant: the system operates at full thermal recovery efficiency at every stage of the bootstrap, with each metallurgical improvement reducing maintenance burden without altering the fundamental architecture.

6. Performance Specifications

Thermal recovery efficiency: greater than 70 percent analytically estimated, design target 80 to 90 percent. Regolith throughput: 50 to 100 kg/hr pilot scale. Operating pressure: 0.2 to 0.3 bar nominal (0.1 to 0.5 bar design range). Operating temperature range: 200 to 900 K across stages (extraction reactor operates separately at higher temperatures). Number of stages: 5 (optimized for recovery efficiency versus system mass). Temperature step per stage: approximately 140 K. Gas composition: H₂, He, CO, CO₂, N₂ from regolith volatiles (CO-enriched composition preferred for higher molecular weight, improved fluidization characteristics, and in-situ iron shot carburization). Gas inventory: approximately 15 grams at 0.2 bar (envelope volume approximately 0.2 m³ at average gas temperature of 600 K). Gas generation rate: approximately 20 to 30 g/hr at 100 kg/hr throughput (from 200 to 300 ppm total recoverable volatiles including mineral decomposition products). Gas leakage rate: approximately 0.4 g/hr (conservative, Knudsen flow regime). Generation-to-loss ratio: greater than 50:1.

Total parasitic power consumption during steady-state operation is estimated at 80 to 250 watts depending on throughput and operating mode. The five stages are connected in parallel to gas supply and return manifolds, so the blower overcomes the pressure drop across one stage rather than all five in series. Per-stage pressure drop is approximately 7,000 Pa at 3 times U_{mf} operating velocity, comprising approximately 6,300 Pa across the sintered metal distributor plate (Ergun equation: 25 micrometer pores, 0.30 porosity, 4 mm thickness) and approximately 400 Pa across the fluidized bed (bed weight per unit area at lunar gravity). Total gas flow through all five stages in parallel is approximately 0.009 m³/s (five stages at 0.0018 m³/s each, based on 0.1 m² bed cross-section and 18 mm/s superficial velocity). Ideal blower power is therefore 7,000 Pa times 0.009 m³/s equals approximately 63 watts, or approximately 125 watts at 50 percent blower efficiency. Maximum system pressure at the blower outlet is approximately 0.27 bar, well within the 0.1 to 0.5 bar design envelope. The remaining parasitic power includes EDS electrodes (5 to 15 watts total), airlock actuators (20 to 40 watts intermittent), sensors and control electronics (10 watts continuous), and the standby blower at 0 watts during normal operation.

System mass budget: pressure vessel (Ti-6Al-4V), 80 kg; alumina ceramic liners (greater than 99.9 percent purity), 40 kg; sintered Inconel distributor plates (8 units), 15 kg; iron shot thermal mass (initial seed stock), 50 kg; EDS electrode arrays (8 m²), 5 kg; gas blower primary and redundant standby, 15 kg; rotary airlocks with ceramic liners (4 units), 20 kg; cyclone separators, 15 kg; cold trap/condenser for water vapor management, 2 kg; multilayer insulation (exterior), 3 kg; sensors and control electronics, 8 kg; structural frame, 25 kg; pre-charge gas bottle, 0.1 kg; iron shot replacement hopper and feed mechanism, 5 kg; gas manifold piping and inter-stage

connections, 80 kg; miscellaneous fasteners and seals, 5 kg; integration hardware and mass contingency (30 percent), 100 kg. Total system mass: approximately 470 kg.

Iron shot annual consumption: 26 to 53 kg/year (pure iron, Stage 1); 13 to 26 kg/year (pearlitic iron, Stage 2); 5 to 15 kg/year (carburized pearlitic iron, Stage 3). MRE iron production capacity: approximately 10 kg/hr. Replacement margin: greater than 1,000:1 at Stage 3.

Sensible heat savings at 100 kg/hr pilot-scale throughput: approximately 12 to 16 kW continuous (calculated as mass flow rate times specific heat capacity times temperature gain: $0.0278 \text{ kg/s} \times 800 \text{ J/kg/K} \times 560 \text{ K}$ equals 12,400 W at 80 percent effectiveness). For MRE operations at 1600 K extraction temperature, total system savings including reduced reactor thermal losses from receiving preheated feed at 760 K instead of 200 K are approximately 25 kW. For hydrogen reduction at 900 K, total savings are approximately 17 kW. Solar array mass equivalence depends on mission power architecture: at 250 W/kg for photovoltaic panels alone, the MRE savings correspond to approximately 100 kg; when energy storage for continuous operations through the 14-day lunar night is included, the effective specific power of the complete power system drops to approximately 25 to 50 W/kg, increasing the avoided power system mass to approximately 500 to 1000 kg. At pilot scale, the system mass of approximately 470 kg achieves positive mass payback when continuous-operation energy storage requirements are considered. Savings scale linearly with throughput: at production-scale throughputs of 1,000 kg/hr or greater, the savings exceed 120 kW sensible heat and the mass payback is strongly positive regardless of power system architecture.

7. Control System and Fault Management

7.1 Blower Control and Redundancy

The primary blower 132a operates continuously during steady-state processing. The standby blower 132b is maintained in ready state with periodic health checks (RPM spin-up test every 24 hours). Automatic crossover from primary to standby occurs within 2 seconds upon detection of RPM deviation exceeding 10 percent from setpoint, differential pressure drop below 80 percent of nominal across the blower, or vibration signature exceeding baseline by 3 sigma. The crossover is managed by the control system without operator intervention.

7.2 EDS Monitoring and Protection

Each EDS electrode phase is equipped with ground fault detection that continuously monitors leakage current through the alumina dielectric. An increase in leakage current above baseline indicates liner thinning or micro-crack formation. If leakage current on any phase exceeds a configurable threshold (set during Phase 2 commissioning based on empirical Paschen characterization of the specific gas composition), that phase is automatically de-energized to prevent micro-arcing. Adjacent phases continue operating, providing partial dust mitigation coverage while the affected zone relies on the pressurized gas environment and fluidized bed mixing for dust management until the next scheduled maintenance interval.

7.3 Safe Mode Choreography

Upon loss of both blowers (dual failure), the system executes a deterministic safe mode sequence. First, regolith feed is halted immediately by stopping the inlet airlock rotation. Second, piezoelectric vibration actuators on each stage vessel are activated to maintain minimal bed agitation and prevent sintering during cooldown. Third, a controlled thermal ramp-down is initiated by reducing or halting the hot regolith return feed. Fourth, the passive thermosiphon effect provides supplementary gas mixing that slows thermal gradient collapse, extending the fault recovery window from minutes to tens of minutes. This does not achieve fluidization but reduces the rate of thermal stratification and localized sintering. The system can be safely parked in this state for extended periods without permanent damage, allowing time for ground-commanded recovery actions.

8. Novelty and Prior Art Differentiation

An extensive review of published literature, patent databases, and NASA technical reports has identified no prior art that anticipates the present invention or renders it obvious.

NASA explored fluidized beds for hydrogen reduction of lunar regolith in the ROxygen project, which used auger stirring and hydrogen gas fluidization within a reaction chamber. The ROxygen Phase I system suffered severe mechanical wear on auger flights from abrasive simulant. The Phase II design transitioned to pneumatic regolith transfer and gas fluidization with a batch-mode pre-heating jacket, but lacked continuous counter-flow heat exchange. No pressurized volatile envelope, no iron shot thermal mass, no EDS integration, and no fines management system were present in either phase.

NASA's Counterflow Regolith Heat Exchanger (CoRHE), developed from 2008 to 2014, used a tube-in-tube concentric design with dual mechanical augers for counter-flow heat exchange. CoRHE achieved approximately 80 percent recovery in ideal simulant testing but was fatally crippled by friction, localized abrasion, and catastrophic jamming when subjected to realistic lunar dust conditions. The heavy, high-torque auger motors consumed significant parasitic power and added substantial payload mass. The present invention specifically addresses the failure modes that limited CoRHE by using fluidized beds (anti-clogging), pressurized gas (convective enhancement), and EDS (anti-fouling), none of which were present in CoRHE.

Various heat pipe and radiation shield concepts have been proposed for extraterrestrial thermal management, but operate on fundamentally different physical principles with recovery efficiencies limited to 50 to 60 percent.

Terrestrial fluidized bed heat exchangers are well-known in industrial mineral processing, but none have been designed for vacuum environments, reduced gravity, electrostatically charged granular material, or integration with extraterrestrial extraction reactors. No prior art discloses in-situ carburization of thermal mass media by process-derived volatile gases within a fluidized bed heat exchanger.

The present invention is novel in the combination of: counter-flow staged fluidized beds as the primary heat recovery mechanism for ISRU regolith; a pressurized processing envelope using process-derived volatile gases to enable convection in vacuum; iron-based thermal mass from extraction byproduct with passive in-situ carburization creating self-hardening closed-loop material synergy; Electrodynamics Dust Shield integration within high-temperature fluidized bed stages with high-purity alumina dielectric isolation; low-power forced gas circulation with N+1 blower redundancy and passive thermosiphon limp-home mixing; active particle size

management addressing the bimodal Geldart A/C transition characteristic of extraterrestrial regolith; and staged bootstrap deployment enabling progressive transition from Earth-supplied to locally-produced and metallurgically optimized thermal mass media. Each feature addresses a specific challenge, and their combination produces a system that simultaneously solves the vacuum, abrasion, electrostatic, clogging, cohesion, and energy-waste problems that have individually or collectively defeated prior approaches.

9. Claims

What is claimed is:

1. A heat recovery system for processing granular material in a reduced-gravity vacuum environment, comprising: a pressurized processing envelope maintained at a pressure between 0.05 and 1.0 bar by gases derived from the granular material being processed; a plurality of fluidized bed stages arranged in counter-current configuration within said envelope, wherein a first granular stream flows through said stages in a first direction and a second granular stream flows through said stages in a second direction opposite to said first direction; and a gas circulation system configured to fluidize said granular material within each stage using said derived gases as the fluidization medium.
2. The system of Claim 1, wherein said granular material comprises extraterrestrial regolith including lunar regolith, Mars regolith, or asteroidal regolith.
3. The system of Claim 1, wherein said derived gases comprise one or more of H₂, He, CO, CO₂, and N₂ extracted from said granular material during thermal processing.
4. The system of Claim 1, further comprising a thermal mass within each fluidized bed stage, said thermal mass comprising metallic particles of 1 to 10 mm diameter.
5. The system of Claim 4, wherein said metallic particles comprise iron or iron alloy produced as a byproduct of Molten Regolith Electrolysis of said granular material.
6. The system of Claim 1, further comprising Electrodynamic Dust Shield electrode arrays on interior surfaces of said fluidized bed stages, configured to prevent accumulation of electrostatically charged fine particles on said surfaces, wherein said electrode arrays are insulated by alumina ceramic liners of greater than 99.9 percent purity excluding transition metal dopants.
7. The system of Claim 1, further comprising a circulation driver comprising a blower or pump configured to maintain superficial gas velocity through said stages at or above the minimum fluidization velocity of said granular material, wherein natural convection arising from the temperature gradient across said stages may provide supplemental gas mixing.
8. The system of Claim 1, further comprising one or more rotary airlock assemblies for transferring said granular material between the vacuum exterior and the pressurized interior, each airlock comprising a gas purge system for pressure equalization and dust containment.
9. The system of Claim 1, wherein said plurality of fluidized bed stages comprises between 3 and 8 stages, each stage maintaining a temperature differential of 80 to 200 K from adjacent stages.

- 10.** The system of Claim 1, wherein the pressurized processing envelope is maintained at between 0.1 and 0.5 bar using gases generated from said granular material at a generation-to-loss ratio exceeding 10:1.
- 11.** The system of Claim 4, wherein the tumbling action of said thermal mass within the fluidized bed disrupts incipient sintering bonds between regolith particles at temperatures approaching 900 to 1000 K.
- 12.** The system of Claim 1, further comprising sacrificial ceramic liners on interior surfaces of said fluidized bed stages and said pressurized envelope, said liners having a Mohs hardness exceeding that of said granular material.
- 13.** The system of Claim 1, wherein said fluidized bed stages each comprise a sintered metal distributor plate with pore size between 10 and 50 micrometers, configured such that the distributor pressure drop exceeds 20 percent of the total per-stage pressure drop to enforce uniform gas distribution across the bed cross-section.
- 14.** The system of Claim 1, wherein said system recovers greater than 70 percent of thermal energy from said second granular stream and transfers it to said first granular stream.
- 15.** The system of Claim 1, wherein the total parasitic power consumption is less than 2 percent of the thermal energy being recovered.
- 16.** A method of recovering thermal energy from processed extraterrestrial regolith, comprising: establishing a pressurized gas environment within a processing envelope using volatile gases extracted from said regolith; passing cold incoming regolith through a series of fluidized bed stages in a first direction; simultaneously passing hot processed regolith through said stages in a counter-current direction; fluidizing said regolith in each stage using said extracted volatile gases as the fluidization medium; and recovering greater than 70 percent of the thermal energy differential between said hot processed regolith and said cold incoming regolith.
- 17.** The method of Claim 16, further comprising using metallic particles produced as a byproduct of said regolith processing as thermal mass within each fluidized bed stage.
- 18.** The method of Claim 16, further comprising operating Electrodynamic Dust Shields on interior surfaces of said fluidized bed stages to prevent fouling by electrostatically charged fine particles.
- 19.** The method of Claim 16, wherein the volatile gas inventory within said pressurized envelope is self-sustaining through continuous liberation of gases from said regolith during thermal processing.

20. An in-situ resource utilization system comprising: an extraction reactor configured to process extraterrestrial regolith at temperatures between 700 and 1900 K; and a heat recovery subsystem coupled to said extraction reactor in a closed-loop material and energy relationship, said heat recovery subsystem comprising a pressurized envelope containing a plurality of counter-current fluidized bed stages that preheat incoming regolith using thermal energy recovered from spent regolith discharged by said extraction reactor, wherein said pressurized envelope is maintained by volatile gases liberated from said regolith during processing and wherein byproducts of said extraction reactor serve as functional components within said heat recovery subsystem.

21. The system of Claim 20, wherein said extraction reactor is a Molten Regolith Electrolysis reactor and wherein metallic iron produced by said reactor is used as thermal mass within said fluidized bed stages.

22. The system of Claim 20, wherein said extraction reactor is a hydrogen reduction reactor and wherein hydrogen gas used in said reactor also serves as a component of the fluidization gas within said fluidized bed stages.

23. The system of Claim 20, further comprising a circulation driver comprising a blower or pump rated at 70 to 150 watts configured to maintain superficial gas velocity at 3 to 5 times the minimum fluidization velocity of the regolith across stages connected in parallel to gas supply and return manifolds, and a redundant standby circulation driver for fault tolerance, wherein natural convection may provide supplemental gas mixing during circulation driver off-states.

24. The system of Claim 1, further comprising one or more cyclone separators in the gas circulation path configured to capture elutriated fine particles from the gas stream and return said particles to one or more of said fluidized bed stages.

25. The system of Claim 24, wherein said cyclone separators capture particles below a cutoff diameter determined by the terminal velocity of said particles at the operating gas velocity, pressure, and gravity of the deployment environment.

26. The system of Claim 1, further comprising a pre-classification stage upstream of said fluidized bed stages configured to remove particles below 40 to 50 micrometers prior to entry into the fluidized beds.

27. The system of Claim 1, wherein the gas composition within said pressurized envelope is enriched in CO relative to the native volatile composition of said granular material, to increase the molecular weight and density of the fluidization gas for improved fluidization characteristics and to provide a carburizing atmosphere for in-situ surface hardening of iron-based thermal mass.

28. The system of Claim 1, further comprising a pre-charge gas vessel containing compressed gas sufficient to establish initial pressurization of said envelope prior to the onset of volatile gas generation from regolith processing.

29. The system of Claim 5, wherein said metallic thermal mass is initially provided as Earth-manufactured seed stock and progressively replaced by iron or iron alloy particles produced on-site from said Molten Regolith Electrolysis, wherein the metallurgical properties of said on-site produced particles are progressively improved through controlled carbon dissolution during electrolysis and in-situ surface carburization by exposure to carbon monoxide within said pressurized envelope.

30. The system of Claim 5, wherein said metallic particles develop a surface layer of iron carbide through exposure to the carbon monoxide component of said derived gases at operating temperatures within said fluidized bed stages, said iron carbide surface layer having a hardness of 800 to 1000 HV and reducing the abrasive-to-metal hardness ratio against silicate regolith minerals to at or below the critical transition threshold of approximately 1.2.

31. The system of Claim 4, further comprising a magnetic separation stage configured to capture ferromagnetic iron fines generated by abrasive degradation of said thermal mass and to separate said iron fines from non-magnetic regolith particles in the discharge stream.

10. Risk Assessment

Abrasion of internal components presents a medium likelihood and medium impact risk. The primary mitigation is sacrificial alumina ceramic liners (Mohs 9 versus regolith Mohs 5-7) with planned replacement intervals. The fluidized bed operation distributes wear uniformly rather than concentrating it in directional flow paths. Iron shot thermal mass absorbs particle-on-particle impact energy that would otherwise concentrate on vessel walls.

Gas envelope seal integrity presents a low to medium likelihood and medium impact risk. The 50:1 generation-to-loss margin, grounded in Knudsen flow regime analysis for vacuum-side leakage, provides substantial headroom for seal degradation. Rotary airlock seals are the most vulnerable component and are designed with ceramic contact surfaces and positive gas purge to minimize regolith ingress into seal interfaces. PTFE transfer film formation may stabilize long-term seal wear rates.

EDS long-duration reliability presents a low to medium likelihood and medium impact risk. Mitigations include high-purity (greater than 99.9 percent) alumina dielectric excluding transition metal dopants, redundant electrode arrays, ground fault detection on each EDS phase for continuous liner integrity monitoring, and the supplementary dust mitigation provided by the pressurized envelope and fluidization. The EDS operating voltage must be validated against the Paschen breakdown curve for the specific gas composition and operating pressure during Phase 2 testing.

Cohesive fines behavior presents a medium to high likelihood and medium impact risk. The Geldart A/C transition character of extraterrestrial regolith means that fine particles may form agglomerates that resist fluidization. This risk is directly addressed by the three-pronged fines management approach (cyclone separation, optional pre-classification, and acceptance of fines loss), the EDS system that disrupts electrostatic agglomeration, the iron shot tumbling action that mechanically shatters fairy castle agglomerates, and the pressurized gas environment that suppresses electrostatic cohesion. Phase 1 modeling will quantify the residual risk.

Blower failure presents a low likelihood and high impact risk. The N+1 redundant blower architecture with automatic crossover within 2 seconds is the primary mitigation. The safe mode choreography (halt feed, activate vibration, thermal ramp-down) provides deterministic fault management. The passive thermosiphon effect extends the fault recovery window by slowing thermal gradient collapse. Blower motors use ceramic bearings and brushless design for longevity.

Iron shot degradation presents a high likelihood and low impact risk. The iron thermal mass will continuously degrade through abrasive wear. This is an expected and designed-for condition, not a failure mode. The MRE byproduct stream provides replacement iron at approximately 1,000 times the consumption rate. In-situ carburization progressively reduces the wear rate over the first hundreds of hours of operation. Iron fines from degradation are captured by cyclone separators and optional magnetic separation.

Rotary airlock seal wear presents a medium likelihood and medium impact risk. The seal wear problem is fundamentally the same challenge that defeated Apollo dust brushes, though PTFE transfer film formation may provide self-stabilizing tribological behavior. Gas purge assists, but the architecture may evolve to screw conveyors or diaphragm valves during Phase 2 bench testing. This is acknowledged and accepted as a maturation item.

11. Development Roadmap

Phase 1 (CFD-DEM Modeling): Computational validation of fluidized bed behavior with realistic extraterrestrial regolith particle size distributions at reduced gravity and low gas pressure. Key deliverables include bubble stability mapping, entrainment rate quantification, fines agglomeration threshold identification, distributor pressure drop validation, electrostatic force modeling at 0.1 to 0.3 bar, and cyclone separation efficiency modeling with Cunningham slip correction for sub-10 micrometer particles at 0.2 bar to determine whether secondary electrostatic precipitation is required for the ultrafine fraction. Thermosiphon loop analysis is complete and confirms blower-driven circulation as the required architecture. Target Technology Readiness Level: 3 to 4.

Phase 2 (Bench-Scale Testing): Laboratory testing with lunar regolith simulant (JSC-1A, LHS-1, or equivalent) in a reduced-pressure chamber. Single-stage fluidized bed at 0.2 bar with instrumented temperature profiles, entrainment measurement, and EDS electrode performance characterization including high-temperature dielectric stability validation of greater than 99.9 percent purity alumina coatings, Frenkel-Poole leakage current measurement at 1000 K, and Paschen breakdown characterization for the specific gas composition and pressure. Iron shot wear rate measurement with regolith simulant, including carburization rate quantification in CO atmosphere. Cyclone separator performance at low pressure with sub-40 micrometer simulant fines. Rotary airlock seal wear testing with abrasive simulant beyond 10,000 cycles to establish realistic degradation curves. Target Technology Readiness Level: 4 to 5.

Phase 3 (Integrated Prototype): Multi-stage prototype with full counter-flow operation, integrated gas circulation, and fines management. Testing under simulated reduced gravity (parabolic flight or drop tower for transient validation). Integration testing with a representative extraction reactor (hydrogen reduction or MRE simulator) including validation of the iron shot closed-loop replenishment cycle and carburization hardening progression. Target Technology Readiness Level: 5 to 6.

12. Abstract of the Disclosure

A multi-stage counter-flow fluidized bed heat exchange system for recovering thermal energy from processed extraterrestrial regolith in reduced-gravity vacuum environments. The system operates within a pressurized envelope maintained at 0.1 to 0.5 bar by volatile gases extracted from the regolith during processing, enabling convective heat transfer impossible in vacuum. Five fluidized bed stages in counter-current configuration achieve an analytically estimated greater than 70 percent thermal energy recovery (design target 80 to 90 percent) between incoming cold regolith and outgoing hot spent regolith over the 200 to 900 K operating range. Iron-based thermal mass from extraction byproduct buffers thermal energy, disrupts sintering, and undergoes passive in-situ surface carburization by carbon monoxide in the gas envelope, progressively hardening from 200 HV to 800 to 1000 HV during normal operation. Electrodynamic Dust Shields on interior surfaces insulated by high-purity alumina prevent fouling by electrostatically charged fines. Forced gas circulation via low-power blower (70 to 150 watts) drives fluidization through parallel-manifold connected stages, with passive thermosiphon providing supplementary mixing during fault states. Active fines management addresses the Geldart A/C transition character of extraterrestrial regolith through cyclone separation, pre-classification, or controlled elutriation. A staged bootstrap deployment transitions the iron thermal mass from Earth-supplied seed stock to locally-produced and progressively optimized media. The system saves 12 to 16 kW of sensible heating power at 100 kg/hr pilot-scale throughput (approximately 25 kW total system savings for MRE operations), with savings scaling linearly to production throughputs.

Declaration

I, Nicholas Dean Perry, declare that I am the sole inventor of the subject matter disclosed in this provisional patent application. All inventive concepts, system architecture decisions, and engineering analysis originated with me.

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